

Quote precisely theorem/result from class you are using.
if using HA problem, prove it and use.
Justify your steps. Total marks: 50.

1. (a). I toss a coin – chance of heads p – till third head appears and then stop. Let X be the number of tosses. Let Y be the time (toss-number) of appearance of second head. Find conditional distribution of Y given $X = 50$. [5]

(b). A rat is stuck in a maze with four doors. Door 1 leads it to safety after 5 minutes of travel. Door 2 leads it back to the maze after 3 minutes of travel. Door 3 leads it back to the maze after 4 minutes of travel. Door 4 leads it to safety after 10 minutes of travel. When in the maze, the rat immediately chooses one of the doors at random. Find the expected time to safety for the rat. [5]
2. (a). Joint density of (X, Y) is given by $f(x, y) = \frac{e^{-y}}{y}$ for $0 < x < y < \infty$. Calculate $E(X^2 | Y = 39)$. [5]

(b). In the usual one dimensional simple symmetric random walk, calculate $\lim_n p_{0,50}^{(n)}$. [5]
3. I have L black balls and L white balls. These are placed in two urns, L balls in each. Every minute, one ball is drawn at random from each of the two urns simultaneously and put into the other urn. Consider the markov chain:
 X_n = the number of black balls in urn 1 after n exchanges.
Show that the chain is irreducible and aperiodic.
Show that the stationary distribution is given by [10]

$$\pi_k = \binom{L}{k}^2 / \binom{2L}{L} \quad k = 0, 1, \dots, L.$$

4. Consider markov chain with six states and transition matrix as below.

$$\begin{array}{c}
 \begin{array}{c} & A & B & C & D & E & F \end{array} \\
 \begin{array}{c} A \\ B \\ C \\ D \\ E \\ F \end{array} \begin{bmatrix} 1/3 & 0 & 0 & 2/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1/2 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 2/3 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 1/2 \\ 0 & 1/2 & 0 & 0 & 1/2 & 0 \end{bmatrix}
 \end{array}$$

Explain communicating classes and period of each class.

If you started from B what happens in the long run?

Calculate $p_{CA}^{(2)}$; $p_{AF}^{(10)}$. [10]

5. I roll a usual fair die once. If I get odd face i , I pick a number in the interval $(i - 2, i + 1)$ at random. If I get an even face j , I pick one number at random from among the two numbers $\{j - 1, j + 1\}$. Let X be the number so selected. Find the distribution function F of X .

[10]

RELAX ENJOY STAY SAFE